Controlled Class-Imbalance Construction for TCGA-UT

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Abstract

Class imbalance in computational pathology is routinely studied under real distributions, where prevalence, class separability, and data-collection history are entangled. This report defines a controlled TCGA-UT benchmark that varies only the imbalance severity of the training distribution while holding validation and test splits fixed. Power-law training distributions are generated under native-prevalence class ordering at three imbalance levels that bracket and extend the native TCGA-UT imbalance ratio, ranging from milder than native to substantially more skewed, over a shared strictly feasible slide pool. Using frozen Virchow2 patch and WSI-bag features with a fixed evidence budget, the same method families evaluated in a prior class-imbalance study on TCGA-UT (Queisler, 2026) are compared across the three constructed severities, including a synthetic patch-augmentation baseline. Method rankings are largely preserved across severities while absolute performance declines monotonically as rare-class support diminishes. At the patch level, synthetic augmentation is competitive at mild imbalance but undergoes the steepest severity-induced decline; OKO is the most robust. WSI-bag methods outperform patch methods throughout, and bag-composition mechanisms—notably RankMix—are the most severity-robust.

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1 Introduction

A prior class-imbalance study on TCGA-UT compared mitigation methods under the dataset's native, moderately imbalanced slide distribution (head-to-tail ≈28:1) across 32 cancer types (Queisler, 2026). Native distributions are realistic but confound several factors simultaneously: how often each class appears, tumor morphology, tissue diversity, slide acquisition, and the representation geometry of the frozen encoder. A controlled construction isolates one factor—training-support frequency—by imposing known power-law distributions while holding validation and test data fixed.

TCGA-UT is a suitable test bed for this isolation because it provides slide labels and patch-level evidence from diagnostic whole-slide images in The Cancer Genome Atlas (Tomczak et al., 2015; Komura et al., 2022). Controlling only the imbalance severity under a fixed native-prevalence class ordering allows the question to be posed cleanly: do mitigation methods maintain their relative advantage as the imbalance between frequent and rare cancer types intensifies along the same ranking?

The report evaluates the same method families as Queisler (2026) under three constructed training distributions that vary only $\lambda \in \{0.5, 1.0, 1.5\}$, the imbalance-severity parameter, while holding the class ranking fixed. A single head-to-tail ratio is only a coarse summary of imbalance; λ instead governs the full shape of the power-law training profile, so every rank—not just the extremes—changes in a controlled way. Patch-level methods include cross-entropy, class weighting, balanced sampling, focal loss, Scholz-style metric losses, CFAL, OKO set learning, and ProGAN synthetic augmentation (Scholz et al., 2024). WSI-bag methods include cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Dataset Construction

The construction shares the TCGA-UT case-disjoint split protocol with Queisler (2026). For each of three random seeds, the validation and test splits are held fixed, and artificial imbalance is applied only to the training split. The assignment is patient-disjoint: all slides and feature rows associated with one TCGA participant remain in exactly one partition. This isolates the effect of training-support frequency: model selection and evaluation are comparable across imbalance parameters and mitigation methods without patient leakage.

All 32 cancer types are retained. Let $C = 32$, let $\mathcal{D}_{\text{train}}$ denote the seed-specific training slide pool, and let a_c be the number of available training slides for class c. The constructed training split must satisfy

$$1 \leq n_c \leq a_c, \qquad \sum_{c=1}^{C} n_c = N, \qquad c \in \{1, \ldots, C\}. \tag{1}$$

Every class receives at least one slide, no class is oversampled beyond its available pool, and the total training set is fixed at N slides shared across all imbalance parameters and seeds.

After slide selection, the input evidence is capped. Patch classifiers use up to 30 deterministic patches per slide. WSI-bag classifiers use up to 30 deterministic frozen feature instances per slide. In both regimes, Virchow2 features remain frozen (Vorontsov et al., 2024; Zimmermann et al., 2024), represented as the concatenation of the CLS token and the mean of the patch tokens (2,560-dimensional vectors). The cap aligns the input budget across regimes and prevents slides with more extracted tumor patches from dominating the training objective.

2.1 Power-Law Training Support

For an ordered class list $(c_1, \ldots, c_C)$, the intended share for rank i under imbalance parameter λ is

$$p_i(\lambda) = \frac{i^{-\lambda}}{\sum_{j=1}^{C} j^{-\lambda}}. \tag{2}$$

When $\lambda = 0$ the intended distribution is uniform over ranks; as λ increases, early-ranked classes receive more training support and late-ranked classes receive less. All constructed splits use the *native-prevalence order*: classes are ranked by their TCGA-UT training-split slide count, so rank 1 is the most frequent cancer type and rank $C = 32$ the least frequent. The power-law form is imposed to obtain controlled, monotonic severity steps; it is not a fit to the native TCGA-UT distribution, which is only moderately imbalanced (≈28:1) and not well described by a single power law.

Raw targets $N\, p_i(\lambda)$ are converted to integer counts by flooring and assigning residual slides by largest fractional remainder. Each class first receives one guaranteed slide; the remaining $N - C$ slides are then allocated according to Equation (2). A regime is accepted only when every rounded target satisfies Equation (1).

2.2 Feasible Pool Size and Realized Profiles

The pool size N must be small enough that no class target exceeds its available support across all evaluated imbalance parameters. Define the maximum feasible pool for a given λ and seed as

$$T^*(\lambda) = \min_c \frac{a_c}{p_c(\lambda)}. \tag{3}$$

For $N \leq T^*(\lambda)$, all targets $N\, p_c(\lambda)$ are feasible. Setting $N = \lfloor \min_s T_s^*(0.5) \rfloor$ over the three seeds guarantees exact feasibility at the mildest evaluated imbalance on every seed. The binding regime is $\lambda = 0.5$ rather than the steepest: the flat power-law profile at $\lambda = 0.5$ requires every class—including rare ones with limited available slides—to contribute a non-trivial share, saturating the rarest class's supply at a lower total N than the peaked profiles at $\lambda \geq 1.0$. For the TCGA-UT training splits, this yields $N = 1{,}132$ slides, approximately 18% of the full available training pool.

The three λ values produce head-to-tail ratios of approximately 6:1 ($\lambda = 0.5$), 30:1 ($\lambda = 1.0$), and 122:1 ($\lambda = 1.5$) (Figure 1). The $\lambda = 0.5$ setting places the constructed distribution well below the native TCGA-UT ratio of approximately 28:1; $\lambda = 1.0$ reproduces near-native severity; and $\lambda = 1.5$ is substantially more skewed than anything observed in the native data.

3 Methods

The method taxonomy follows the companion benchmark, but this report uses it to study controlled training distributions rather than only the native TCGA-UT distribution. Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or synthetic augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, prototype-affinity objectives, or OKO set supervision. WSI-specific methods change how slide bags are represented or mixed.

Full method definitions follow Queisler (2026) and Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are WSI-bag methods whose mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025).

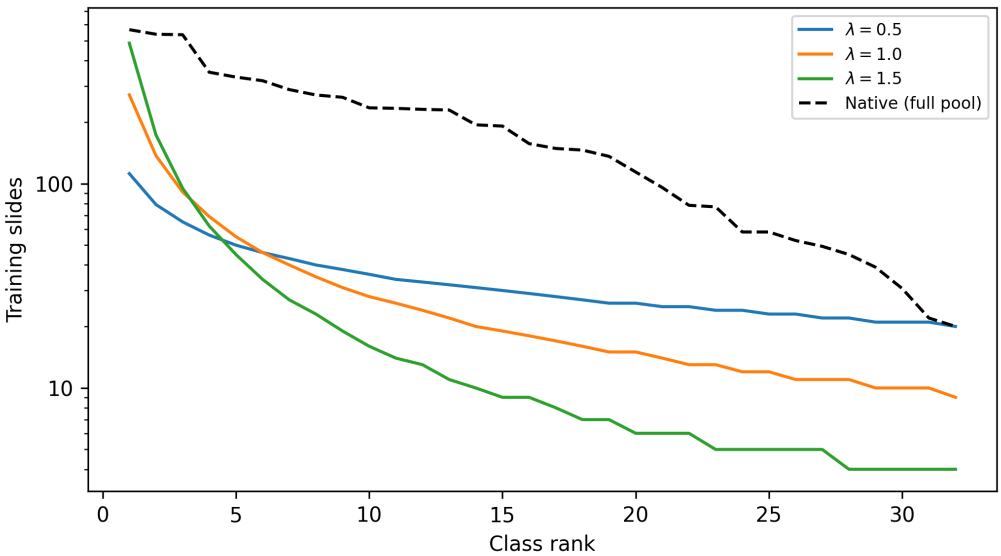

Figure 1: Constructed training support by class rank under the three evaluated imbalance parameters, with the native full-pool distribution shown as a reference. Classes are ordered by native prevalence (rank 1 = most frequent).

Family	Methods	Regime
No-mitigation baseline	CE / MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling–loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 1: Method families evaluated under constructed TCGA-UT training distributions, following the taxonomy of Queisler (2026). Patch and WSI-bag methods are compared only within their own input regime.

4 Evaluation Protocol

Each constructed split is indexed by seed and imbalance parameter $\lambda \in \{0.5, 1.0, 1.5\}$. For every method and constructed regime, hyperparameters are selected by validation macro F1 across three seeds; ties are broken by validation balanced accuracy. Selected configurations are evaluated on the fixed test split, and results are reported as mean $\pm$ standard deviation across seeds. Macro F1 is the primary discrimination metric because it weights cancer types equally. Balanced accuracy and accuracy are reported as complementary discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the case-disjoint splits, frozen Virchow2 representation family, and 30-instance evidence budget, but they differ in prediction unit and aggregation mechanism.

Calibration is evaluated alongside discrimination using negative log likelihood (NLL), multiclass Brier score (Brier, 1950), and expected calibration error (ECE; Naeini et al., 2015; Guo et al., 2017). For each method and seed, a single temperature is fit by minimizing NLL on the validation split and applied unchanged to the test score matrix before softmax (Guo et al., 2017). Tables 4 and 5 report ECE before and after temperature scaling; full NLL and Brier metrics are given in Tables 7 and 8.

5 Results

The analysis asks how methods respond as imbalance severity increases from $\lambda = 0.5$ to $\lambda = 1.5$ under the native-prevalence class order. The realized support profiles are summarised in Table 6 (Appendix A); Tables 2 and 3 compare mitigation methods within the patch-feature and WSI-bag regimes.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		Native	
	BAcc	F1	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.721 ± 0.012	0.693 ± 0.006	0.704 ± 0.012	0.693 ± 0.016	0.634 ± 0.006	0.641 ± 0.009	0.767 ± 0.006	0.774 ± 0.005
Focal loss	0.718 ± 0.011	0.692 ± 0.007	0.698 ± 0.016	0.689 ± 0.018	0.630 ± 0.005	0.641 ± 0.009	0.758 ± 0.007	0.765 ± 0.009
CFAL	0.724 ± 0.009	0.697 ± 0.007	0.710 ± 0.014	0.704 ± 0.015	0.632 ± 0.013	0.645 ± 0.014	0.789 ± 0.007	0.798 ± 0.009
CE + soft F1	0.721 ± 0.012	0.685 ± 0.004	0.713 ± 0.007	0.686 ± 0.014	0.638 ± 0.021	0.630 ± 0.019	0.781 ± 0.002	0.780 ± 0.003
CE + soft MCC	0.719 ± 0.006	0.690 ± 0.009	0.698 ± 0.010	0.684 ± 0.009	0.634 ± 0.019	0.633 ± 0.018	0.772 ± 0.004	0.770 ± 0.005
Balanced sampling	0.719 ± 0.012	0.691 ± 0.007	0.701 ± 0.015	0.688 ± 0.018	0.631 ± 0.012	0.635 ± 0.013	0.769 ± 0.007	0.771 ± 0.014
OKO	0.722 ± 0.012	0.689 ± 0.008	0.712 ± 0.011	0.694 ± 0.018	0.644 ± 0.015	0.648 ± 0.014	0.791 ± 0.005	0.788 ± 0.003
ProGAN augmentation	0.729 ± 0.003	0.698 ± 0.001	0.712 ± 0.003	0.700 ± 0.015	0.624 ± 0.016	0.636 ± 0.016	0.760 ± 0.005	0.766 ± 0.002

Table 2: Patch-feature test results (balanced accuracy and macro F1) for validation-selected configurations under constructed training distributions. The Native column reports the companion full-pool benchmark (Queisler, 2026) (native ≈28:1 imbalance, complete training pool) as a reference; it is not part of the constructed experiment.

Patch-feature results show a consistent, monotone decline as imbalance intensifies (Table 2). ProGAN augmentation leads in balanced accuracy at the mildest severity but reverses to last at the harshest, undergoing the steepest decline of any patch method; OKO and CE+soft-F1 are the most stable and retain the lead at $\lambda = 1.5$. Focal loss is consistently the weakest among the remaining methods. The key observation is one of scale: the spread across methods at any fixed λ is only one to two percentage points, whereas the severity-induced decline from $\lambda = 0.5$ to $\lambda = 1.5$ is roughly eight to eleven percentage points—choosing the right severity matters far more than choosing the best method. The native full-pool benchmark (rightmost block) sits above even the mildest constructed severity, despite carrying a larger imbalance (≈28:1) than $\lambda = 0.5$ (≈6:1); because it draws on the complete training pool rather than the constructed subset, this gap isolates training-set size—not imbalance—as the dominant lever on absolute patch performance.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		Native	
	BAcc	F1	BAcc	F1	BAcc	F1	BAcc	F1
Weighted CE	0.823 ± 0.019	0.793 ± 0.017	0.799 ± 0.008	0.787 ± 0.007	0.736 ± 0.027	0.731 ± 0.021	0.833 ± 0.019	0.835 ± 0.011
Focal loss	0.821 ± 0.013	0.792 ± 0.012	0.801 ± 0.009	0.792 ± 0.007	0.703 ± 0.029	0.708 ± 0.031	0.823 ± 0.019	0.824 ± 0.008
Balanced sampling	0.809 ± 0.014	0.776 ± 0.010	0.797 ± 0.010	0.785 ± 0.018	0.724 ± 0.042	0.720 ± 0.040	0.829 ± 0.010	0.831 ± 0.009
RankMix	0.830 ± 0.016	0.793 ± 0.015	0.817 ± 0.024	0.795 ± 0.020	0.765 ± 0.029	0.751 ± 0.019	0.864 ± 0.002	0.854 ± 0.008
SC-MIL	0.813 ± 0.015	0.781 ± 0.014	0.800 ± 0.013	0.787 ± 0.008	0.745 ± 0.022	0.743 ± 0.015	0.837 ± 0.013	0.839 ± 0.008
MDE-MIL	0.836 ± 0.013	0.805 ± 0.012	0.825 ± 0.009	0.814 ± 0.007	0.730 ± 0.009	0.731 ± 0.014	0.849 ± 0.014	0.865 ± 0.011

Table 3: WSI-bag test results (balanced accuracy and macro F1) for validation-selected configurations under constructed training distributions. The Native column is the companion full-pool benchmark (Queisler, 2026) reference, as in Table 2.

The WSI-bag regime shows higher absolute performance than the patch regime (Table 3), roughly ten balanced-accuracy points above patch methods at equivalent imbalance. MDE-MIL leads at mild and moderate severities, but RankMix overtakes it at the harshest regime and is the most severity-robust method overall, while focal loss is the least stable. MDE-MIL's advantage over simple loss weighting erodes at $\lambda = 1.5$, where it falls narrowly below Weighted CE—consistency regularisation provides diminishing returns as rare-class support becomes critically scarce. The ordering of bag-composition methods (RankMix, SC-MIL) over loss-based variants holds across all three regimes, confirming that WSI-specific aggregation mechanisms retain their advantage under increasing imbalance. As in the patch regime, the native full-pool benchmark (rightmost block) exceeds even $\lambda = 0.5$ despite its larger imbalance, again pointing to training-set size rather than imbalance severity as the main driver of absolute performance.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		T
	ECE	ECE+TS	ECE	ECE+TS	ECE	ECE+TS	
Weighted CE	0.156 ± 0.003	0.015 ± 0.002	0.157 ± 0.013	0.016 ± 0.005	0.174 ± 0.008	0.011 ± 0.006	2.43 ± 0.01
Focal loss	0.098 ± 0.003	0.010 ± 0.001	0.100 ± 0.012	0.012 ± 0.002	0.150 ± 0.008	0.013 ± 0.004	1.71 ± 0.22
CFAL	0.723 ± 0.009	0.067 ± 0.007	0.728 ± 0.021	0.070 ± 0.008	0.695 ± 0.013	0.077 ± 0.006	0.05 ± 0.00
CE + soft F1	0.179 ± 0.007	0.016 ± 0.004	0.184 ± 0.012	0.014 ± 0.002	0.217 ± 0.017	0.013 ± 0.003	3.22 ± 0.26
CE + soft MCC	0.162 ± 0.001	0.012 ± 0.004	0.169 ± 0.008	0.013 ± 0.002	0.183 ± 0.019	0.015 ± 0.005	2.51 ± 0.03
Balanced sampling	0.156 ± 0.005	0.016 ± 0.002	0.158 ± 0.013	0.014 ± 0.003	0.182 ± 0.011	0.017 ± 0.004	2.37 ± 0.05
OKO	0.071 ± 0.011	0.030 ± 0.003	0.193 ± 0.006	0.041 ± 0.004	0.192 ± 0.013	0.037 ± 0.007	0.70 ± 0.08
ProGAN augmentation	0.189 ± 0.001	0.014 ± 0.004	0.192 ± 0.017	0.019 ± 0.002	0.224 ± 0.009	0.019 ± 0.006	4.05 ± 0.08

Table 4: Patch-feature calibration: ECE before (raw) and after (TS) temperature scaling across evaluated imbalance parameters. Fitted temperature T is averaged across seeds and severities. Full NLL and Brier metrics appear in Table 7.

Method	$\lambda = 0.5$		$\lambda = 1.0$		$\lambda = 1.5$		T
	ECE	ECE+TS	ECE	ECE+TS	ECE	ECE+TS	
Weighted CE	0.099 ± 0.012	0.022 ± 0.007	0.092 ± 0.010	0.022 ± 0.014	0.109 ± 0.009	0.023 ± 0.003	1.98 ± 0.05
Focal loss	0.049 ± 0.013	0.019 ± 0.004	0.054 ± 0.008	0.016 ± 0.005	0.105 ± 0.020	0.031 ± 0.004	1.46 ± 0.14
Balanced sampling	0.105 ± 0.010	0.025 ± 0.007	0.099 ± 0.012	0.020 ± 0.004	0.124 ± 0.020	0.027 ± 0.003	2.05 ± 0.06
RankMix	0.025 ± 0.015	0.020 ± 0.009	0.026 ± 0.005	0.018 ± 0.005	0.065 ± 0.008	0.026 ± 0.004	1.16 ± 0.15
SC-MIL	0.096 ± 0.006	0.026 ± 0.005	0.093 ± 0.011	0.018 ± 0.004	0.110 ± 0.001	0.024 ± 0.006	1.90 ± 0.04
MDE-MIL	0.051 ± 0.007	0.037 ± 0.005	0.042 ± 0.006	0.034 ± 0.007	0.044 ± 0.004	0.039 ± 0.010	0.96 ± 0.10

Table 5: WSI-bag calibration: ECE and fitted temperature T across evaluated imbalance parameters. Columns as in Table 4.

Probability Calibration. In the patch regime, CFAL is a calibration outlier: its affinity-based outputs are nearly flat over 32 classes, requiring strong temperature concentration ($T \approx 0.05$) and remaining substantially miscalibrated (ECE ≈ 0.07) even after scaling—roughly five times higher than all other patch methods post-scaling. All other patch methods achieve ECE below 0.04 after temperature scaling. OKO shows a non-monotone calibration trajectory: well-calibrated at mild severity (ECE $\approx$ 0.07 raw at $\lambda = 0.5$) but deteriorating to ECE $\approx$ 0.19 at moderate severity before scaling. ProGAN augmentation requires the highest temperature correction of any patch method ($T \approx 4.05$), consistent with the highly peaked, overconfident predictions that synthetic-feature training tends to produce; post-scaling ECE is unremarkable (ECE+TS $\approx$ 0.014–0.019). In the WSI-bag regime, RankMix is already well calibrated before scaling (ECE $\approx$ 0.025); MDE-MIL shows only marginal improvement from temperature scaling and retains residual miscalibration (ECE+TS $\approx$ 0.037–0.039). All WSI-bag methods reach ECE at or below 0.039 after scaling. Calibration rankings are broadly stable across imbalance severities.

6 Discussion

The controlled construction reveals two complementary patterns. First, method rankings are broadly preserved across imbalance severities, with moderate reshuffling among closely performing methods at the extremes. Second, balanced accuracy declines monotonically with λ for almost all methods, with a total range of eight to nine percentage points from the mildest to the strongest constructed regime. These patterns suggest that the relative utility of mitigation methods is largely robust to the particular severity chosen within this range, while absolute performance degrades substantially as rare-class support diminishes.

At the patch level, declines span from approximately 7.8 percentage points for OKO to 10.5 for ProGAN augmentation. ProGAN leads all patch methods at the mildest severity ($\lambda = 0.5$, BAcc 0.729) but drops to last at the harshest ($\lambda = 1.5$, BAcc 0.624)—a reversal that reflects the difficulty of training patch-level generative models when tail-class support falls to single digits: synthetic features complement real training data at mild imbalance but cannot compensate for critically scarce patches at severe imbalance. OKO and CE+soft-F1 maintain their relative advantage at the harshest severity; focal loss is consistently weakest among the remaining methods throughout.

In the WSI-bag regime, RankMix stands out for its stability, declining by only 6.5 percentage points from $\lambda = 0.5$ to $\lambda = 1.5$. Focal MIL is the least stable, dropping by nearly 12 percentage points and landing at the bottom of the WSI rankings at $\lambda = 1.5$. MDE-MIL declines more than RankMix and falls narrowly below Weighted CE at the harshest regime, showing that consistency regularisation provides diminishing returns as rare-class support becomes critically scarce. The ordering of bag-composition methods (RankMix, SC-MIL) over simple loss-based variants holds throughout, confirming that WSI-specific aggregation mechanisms are severity-robust.

Calibration analysis complements discrimination: CFAL leads on patch discrimination but requires the strongest temperature correction and retains the highest post-scaling ECE. In the WSI-bag regime, RankMix is already well calibrated before scaling, while MDE-MIL shows only marginal improvement from temperature scaling despite leading on discrimination at mild and moderate severities.

Taken together, the controlled construction shows that when training-support frequency is imposed directly via a power-law distribution, aggregate discrimination declines predictably but modestly with severity, and the method ranking remains informative across the range tested. The prediction unit—patch versus WSI-bag—continues to be the dominant determinant of absolute performance: WSI-bag methods achieve balanced accuracy approximately ten percentage points higher than their patch-feature counterparts at equivalent imbalance.

7 Conclusion

This report defines a full-scale controlled TCGA-UT imbalance benchmark that keeps all 32 cancer types, preserves patient-disjoint validation and test splits, and varies only the training-support frequency under native-prevalence class ordering at $\lambda \in \{0.5, 1.0, 1.5\}$. The shared pool of 1,132 slides is determined by the mildest parameter $\lambda = 0.5$, whose flat profile saturates the rarest class's available supply earliest. The three regimes span head-to-tail ratios of approximately 6:1, 30:1, and 122:1, bracketing and extending the native TCGA-UT ratio ($\approx$28:1). Method rankings are broadly preserved across all three constructed severities, while balanced accuracy declines by eight to eleven percentage points from the mildest to the strongest regime. At the patch level, ProGAN augmentation leads at mild severity ($\lambda = 0.5$, BAcc 0.729) and undergoes the steepest decline (10.5 pp), reversing from best to worst as tail-class support diminishes; OKO and CE+soft-F1 are most robust at the harshest severity; the within-method spread of approximately one to two percentage points at any fixed severity is substantially smaller than the severity-induced decline. At the WSI-bag level, bag-composition methods (RankMix, SC-MIL) consistently outperform loss-based variants regardless of severity; RankMix is the most severity-robust method, while MDE-MIL's advantage over loss-based baselines erodes under the most extreme imbalance. The prediction unit remains the dominant performance determinant: WSI-bag methods achieve balanced accuracy roughly 12 percentage points above patch-feature methods at equivalent imbalance. Temperature scaling is sufficient to align probability outputs across all methods, though CFAL requires the largest correction in the patch regime and MDE-MIL shows limited benefit from scaling in the WSI-bag regime.

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A Constructed Training-Support Profiles

λ	Training slides	Support range	Head:tail ratio
0.5	1132	20–112	≈6:1
1.0	1132	9–272	≈30:1
1.5	1132	4–487	≈122:1
native (full pool)	6110	20–567	≈28:1

Table 6: Constructed training-support profiles under native-prevalence ordering. Support range and head:tail ratio are the minimum and maximum per-class slide counts averaged across seeds. The native full-pool row is shown for reference; it is not part of the constructed experiment.

B Full Calibration Metrics

Method	λ	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
Balanced sampling	$\lambda = 0.5$	1.314 ± 0.117	0.839 ± 0.039	0.390 ± 0.013	0.339 ± 0.012	0.156 ± 0.005	0.016 ± 0.002	2.441 ± 0.095
Balanced sampling	$\lambda = 1.0$	1.301 ± 0.051	0.841 ± 0.051	0.391 ± 0.033	0.339 ± 0.026	0.158 ± 0.013	0.014 ± 0.003	2.334 ± 0.045
Balanced sampling	$\lambda = 1.5$	1.620 ± 0.112	1.060 ± 0.048	0.457 ± 0.023	0.399 ± 0.013	0.182 ± 0.011	0.017 ± 0.004	2.334 ± 0.045
CE + soft F1	$\lambda = 0.5$	1.626 ± 0.137	0.879 ± 0.044	0.413 ± 0.015	0.343 ± 0.011	0.179 ± 0.007	0.016 ± 0.004	2.884 ± 0.112
CE + soft F1	$\lambda = 1.0$	1.819 ± 0.050	0.889 ± 0.050	0.417 ± 0.032	0.343 ± 0.024	0.184 ± 0.012	0.014 ± 0.002	3.258 ± 0.124
CE + soft F1	$\lambda = 1.5$	2.274 ± 0.170	1.047 ± 0.060	0.486 ± 0.038	0.391 ± 0.029	0.217 ± 0.017	0.013 ± 0.003	3.520 ± 0.067
CE + soft MCC	$\lambda = 0.5$	1.375 ± 0.041	0.846 ± 0.017	0.397 ± 0.009	0.341 ± 0.009	0.162 ± 0.001	0.012 ± 0.004	2.552 ± 0.085
CE + soft MCC	$\lambda = 1.0$	1.392 ± 0.045	0.874 ± 0.039	0.411 ± 0.026	0.353 ± 0.023	0.169 ± 0.008	0.013 ± 0.002	2.496 ± 0.095
CE + soft MCC	$\lambda = 1.5$	1.713 ± 0.148	1.053 ± 0.059	0.451 ± 0.041	0.392 ± 0.030	0.183 ± 0.019	0.015 ± 0.005	2.497 ± 0.128
CFAL	$\lambda = 0.5$	3.232 ± 0.006	0.995 ± 0.042	0.952 ± 0.000	0.352 ± 0.014	0.723 ± 0.009	0.067 ± 0.007	0.050 ± 0.000
CFAL	$\lambda = 1.0$	3.228 ± 0.011	0.997 ± 0.087	0.951 ± 0.001	0.349 ± 0.029	0.728 ± 0.021	0.070 ± 0.008	0.050 ± 0.000
CFAL	$\lambda = 1.5$	3.246 ± 0.002	1.165 ± 0.049	0.953 ± 0.000	0.394 ± 0.018	0.695 ± 0.013	0.077 ± 0.006	0.050 ± 0.000
Focal loss	$\lambda = 0.5$	0.933 ± 0.040	0.819 ± 0.025	0.355 ± 0.010	0.336 ± 0.010	0.098 ± 0.003	0.010 ± 0.001	1.565 ± 0.030
Focal loss	$\lambda = 1.0$	0.933 ± 0.055	0.818 ± 0.053	0.357 ± 0.031	0.337 ± 0.027	0.100 ± 0.012	0.012 ± 0.002	1.548 ± 0.052
Focal loss	$\lambda = 1.5$	1.315 ± 0.076	0.970 ± 0.045	0.417 ± 0.017	0.373 ± 0.015	0.150 ± 0.008	0.013 ± 0.004	2.020 ± 0.000
OKO	$\lambda = 0.5$	0.931 ± 0.029	0.886 ± 0.031	0.352 ± 0.012	0.343 ± 0.012	0.071 ± 0.011	0.030 ± 0.003	0.804 ± 0.016
OKO	$\lambda = 1.0$	1.083 ± 0.076	0.902 ± 0.074	0.383 ± 0.028	0.337 ± 0.029	0.193 ± 0.006	0.041 ± 0.004	0.636 ± 0.012
OKO	$\lambda = 1.5$	1.188 ± 0.040	1.019 ± 0.047	0.410 ± 0.019	0.365 ± 0.022	0.192 ± 0.013	0.037 ± 0.007	0.651 ± 0.000
ProGAN augmentation	$\lambda = 0.5$	2.166 ± 0.069	0.855 ± 0.012	0.414 ± 0.002	0.332 ± 0.002	0.189 ± 0.001	0.014 ± 0.004	3.978 ± 0.077
ProGAN augmentation	$\lambda = 1.0$	2.226 ± 0.230	0.866 ± 0.074	0.419 ± 0.037	0.334 ± 0.027	0.192 ± 0.017	0.019 ± 0.002	4.024 ± 0.157
ProGAN augmentation	$\lambda = 1.5$	2.745 ± 0.185	1.032 ± 0.060	0.485 ± 0.020	0.383 ± 0.019	0.224 ± 0.009	0.019 ± 0.006	4.159 ± 0.080
Weighted CE	$\lambda = 0.5$	1.303 ± 0.074	0.830 ± 0.027	0.387 ± 0.012	0.335 ± 0.011	0.156 ± 0.003	0.015 ± 0.002	2.441 ± 0.095
Weighted CE	$\lambda = 1.0$	1.297 ± 0.078	0.825 ± 0.056	0.387 ± 0.034	0.334 ± 0.028	0.157 ± 0.013	0.016 ± 0.005	2.413 ± 0.047
Weighted CE	$\lambda = 1.5$	1.530 ± 0.086	0.969 ± 0.043	0.431 ± 0.016	0.371 ± 0.015	0.174 ± 0.008	0.011 ± 0.006	2.440 ± 0.047

Table 7: Patch-feature calibration: NLL, Brier score, and ECE before (raw) and after (TS) temperature scaling, with fitted temperature T. One row per method per imbalance parameter.

Method	λ	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
MDE-MIL	$\lambda = 0.5$	0.620 ± 0.021	0.588 ± 0.021	0.226 ± 0.010	0.221 ± 0.010	0.051 ± 0.007	0.037 ± 0.005	0.813 ± 0.032
MDE-MIL	$\lambda = 1.0$	0.582 ± 0.029	0.580 ± 0.029	0.220 ± 0.010	0.220 ± 0.011	0.042 ± 0.006	0.034 ± 0.007	1.049 ± 0.020
MDE-MIL	$\lambda = 1.5$	0.741 ± 0.053	0.740 ± 0.052	0.276 ± 0.013	0.276 ± 0.013	0.044 ± 0.004	0.039 ± 0.010	1.015 ± 0.039
Balanced sampling	$\lambda = 0.5$	0.873 ± 0.054	0.611 ± 0.022	0.271 ± 0.017	0.245 ± 0.012	0.105 ± 0.010	0.025 ± 0.007	1.999 ± 0.101
Balanced sampling	$\lambda = 1.0$	0.850 ± 0.030	0.611 ± 0.030	0.268 ± 0.016	0.246 ± 0.012	0.099 ± 0.012	0.020 ± 0.004	2.021 ± 0.067
Balanced sampling	$\lambda = 1.5$	1.140 ± 0.249	0.790 ± 0.137	0.330 ± 0.050	0.300 ± 0.044	0.124 ± 0.020	0.027 ± 0.003	2.143 ± 0.223
Focal loss	$\lambda = 0.5$	0.602 ± 0.036	0.563 ± 0.026	0.233 ± 0.014	0.228 ± 0.012	0.049 ± 0.013	0.019 ± 0.004	1.310 ± 0.000
Focal loss	$\lambda = 1.0$	0.624 ± 0.062	0.566 ± 0.055	0.233 ± 0.027	0.228 ± 0.025	0.054 ± 0.008	0.016 ± 0.005	1.416 ± 0.027
Focal loss	$\lambda = 1.5$	0.902 ± 0.090	0.744 ± 0.057	0.309 ± 0.034	0.290 ± 0.026	0.105 ± 0.020	0.031 ± 0.004	1.654 ± 0.000
Weighted CE	$\lambda = 0.5$	0.848 ± 0.055	0.600 ± 0.024	0.263 ± 0.018	0.238 ± 0.012	0.099 ± 0.012	0.022 ± 0.007	2.022 ± 0.115
Weighted CE	$\lambda = 1.0$	0.807 ± 0.068	0.582 ± 0.048	0.251 ± 0.025	0.231 ± 0.022	0.092 ± 0.010	0.022 ± 0.014	2.020 ± 0.000
Weighted CE	$\lambda = 1.5$	0.940 ± 0.070	0.708 ± 0.046	0.302 ± 0.024	0.275 ± 0.020	0.109 ± 0.009	0.023 ± 0.003	1.912 ± 0.073
RankMix	$\lambda = 0.5$	0.624 ± 0.045	0.623 ± 0.044	0.240 ± 0.019	0.239 ± 0.018	0.025 ± 0.015	0.020 ± 0.009	1.015 ± 0.052
RankMix	$\lambda = 1.0$	0.573 ± 0.055	0.569 ± 0.051	0.230 ± 0.026	0.229 ± 0.026	0.026 ± 0.005	0.018 ± 0.005	1.097 ± 0.021
RankMix	$\lambda = 1.5$	0.706 ± 0.063	0.654 ± 0.045	0.264 ± 0.016	0.257 ± 0.015	0.065 ± 0.008	0.026 ± 0.004	1.370 ± 0.026
SC-MIL	$\lambda = 0.5$	0.773 ± 0.044	0.577 ± 0.017	0.262 ± 0.010	0.239 ± 0.004	0.096 ± 0.006	0.026 ± 0.005	1.913 ± 0.098
SC-MIL	$\lambda = 1.0$	0.765 ± 0.087	0.586 ± 0.059	0.259 ± 0.031	0.239 ± 0.025	0.093 ± 0.011	0.018 ± 0.004	1.848 ± 0.036
SC-MIL	$\lambda = 1.5$	0.934 ± 0.070	0.693 ± 0.055	0.300 ± 0.012	0.274 ± 0.015	0.110 ± 0.001	0.024 ± 0.006	1.932 ± 0.037

Table 8: WSI-bag calibration: NLL, Brier score, and ECE before and after temperature scaling. Columns as in Table 7.